Improving Binary Image Classification Using Transfer Learning, DCGAN Augmentation, and CycleGAN Style Transfer

ZERROUKI Aghilas
University of Paris-Saclay
Paris, France
20251127@etud.univ-evry.fr

***Abstract*—This paper investigates binary image classification between cows and horses using a severely limited dataset of only 170 images. Small datasets commonly cause overfitting and weak generalization when training convolutional neural networks (CNNs) from scratch. To address this, we progressively explored five strategies: (1) a baseline CNN, (2) a regularised CNN, (3) transfer learning with ImageNet pretrained models, (4) Deep Convolutional Generative Adversarial Networks (DCGAN) for synthetic data augmentation, and (5) Cycle-Consistent Generative Adversarial Networks (CycleGAN) for unpaired domain translation followed by classifier retraining on the translated images. The optimised transfer learning classifier achieved 91.46% test accuracy. Adding CycleGAN-translated images to the training set reduced accuracy to 68.29%, confirming that synthetic image quality on tiny datasets is insufficient to benefit discriminative classifiers. CycleGAN nevertheless successfully transferred visual style between domains while preserving pose and scene geometry, demonstrating its value as a domain-adaptation and visualisation tool rather than a data-augmentation strategy. Our results demonstrate that transfer learning is the most effective strategy for small-dataset classification, while GANs offer complementary value for generation and domain adaptation tasks.**

***Index Terms*—Binary Classification, CNN, Transfer Learning, Data Augmentation, GAN, DCGAN, CycleGAN, Style Transfer, Computer Vision, Small Dataset**

I. Introduction

Image classification is a core task in computer vision with applications ranging from medical imaging to autonomous driving. Deep learning models yield state-of-the-art performance when trained on large, balanced datasets; however, collecting and annotating such data is often prohibitively expensive in real-world scenarios [5], [6].

In this work, the classification targets are two visually similar animal categories — *cow* and *horse* — with only 170 labelled images available in total. This setting introduces several well-known challenges:

- **Overfitting** — the model memorises rather than generalises.
- **High variance** — accuracy fluctuates strongly across runs.
- **Limited diversity** — restricted pose, texture, and background variation.
- **Poor feature learning** — CNNs trained from random initialisation cannot learn rich representations from so few examples.

To overcome these limitations we follow a progressive experimental protocol that combines discriminative models (classifiers) with generative models (GANs), systematically evaluating each strategy. Section II describes the dataset; Sections III–VI detail each method; Section VIII summarises quantitative results; and Section IX draws conclusions and outlines future work.

II. Dataset

TABLE I: Dataset statistics and split.

Property	Value
Total images	170
Training set	82
Test set	82
Number of classes	2 (Cow / Horse)
Class balance	Approximately equal
Image format	RGB, variable resolution

All images were resized to 64×64 pixels for GAN training and to 224×224 pixels for transfer learning models. Standard augmentation (horizontal flipping, random cropping, colour jitter) was applied during classifier training to further increase effective diversity.

III. Baseline and Regularised CNN

A. Baseline Architecture

The first experiments used a shallow CNN trained from random initialisation: three convolutional blocks (Conv–BN–ReLU–MaxPool), followed by a fully-connected classifier with softmax output. The cross-entropy loss is:

$$\mathcal{L}_{\text{CE}} = -\sum_{c=1}^{C} y_c \log \hat{y}_c \tag{1}$$

Training accuracy quickly reached 90%+ while test accuracy stagnated between 55% and 70%, a clear sign of overfitting.

B. Regularised CNN

Adding dropout ($p = 0.5$), L2 weight decay, and increased data augmentation raised test accuracy to approximately 73%, confirming that regularisation alone partially mitigates overfitting but remains insufficient for such a small dataset.

IV. Transfer Learning

A. Motivation

Transfer learning leverages feature representations already learnt from large-scale data (ImageNet, $\sim$1.2M images) and adapts them to the target domain with minimal additional training data [5], [6].

B. Two-Stage Fine-Tuning Protocol

Stage A — Classifier warm-up.
The backbone weights are frozen (θ_{backbone} fixed). Only the newly added classification head is trained for 10 epochs with a learning rate of 10^{-3}. This prevents the randomly initialised head from corrupting pretrained features.

Stage B — Fine-tuning.
The upper backbone layers are unfrozen and the entire network is trained end-to-end with a lower learning rate (10^{-5}) and cosine annealing. Early stopping on validation loss prevents re-overfitting.

Pretrained weights initialise the model:

$$\theta_0 = \theta_{\text{ImageNet}} \tag{2}$$

C. Results

The final transfer learning model achieves **91.46%** test accuracy — a gain of more than 21 percentage points over the regularised CNN baseline. Detailed metrics are reported in Section VIII.

V. DCGAN for Synthetic Image Generation

A. Architecture and Objective

A Deep Convolutional GAN (DCGAN) [2] was trained separately on cow images and horse images to learn the data distribution p_{data} and generate novel synthetic samples.

The minimax adversarial objective is:

$$\min_G \max_D \; \mathbb{E}[\log D(x)]_{x\sim p_{\text{data}}} + \mathbb{E}[\log(1 - D(G(z)))]_{z\sim p_z} \tag{3}$$

where $G(z) : \mathbb{R}^{100} \to \mathbb{R}^{3\times 64\times 64}$ is the generator and $D(x) \in [0, 1]$ is the discriminator's real/fake probability.

Generator — five transposed convolution layers upsample $4\times 4 \to 64\times 64$, each followed by Batch Normalisation and ReLU; the final layer uses $\tanh$ to map outputs to $[-1, 1]$:

$$h_k = \text{ReLU}(\text{BN}(\text{ConvT}(h_{k-1}))), \quad \hat{x} = \tanh(\text{ConvT}(h_K)) \tag{4}$$

Discriminator — mirrors the generator with strided convolutions that halve spatial resolution at each stage, ending with a sigmoid unit.

B. Loss Functions

$$\mathcal{L}_D = -\mathbb{E}[\log D(x)] - \mathbb{E}[\log(1 - D(G(z)))] \tag{5}$$

$$\mathcal{L}_G = -\mathbb{E}[\log D(G(z))] \tag{6}$$

C. Training Behaviour and Results

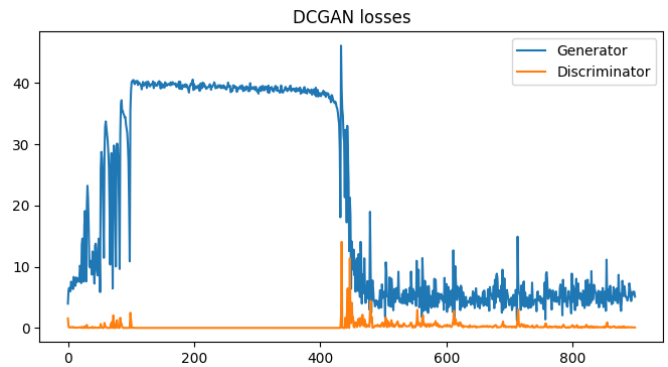

Fig. 1: DCGAN training loss curves. Oscillations reflect the competitive generator–discriminator dynamic; the discriminator initially converges faster, causing temporary instability before both networks stabilise.

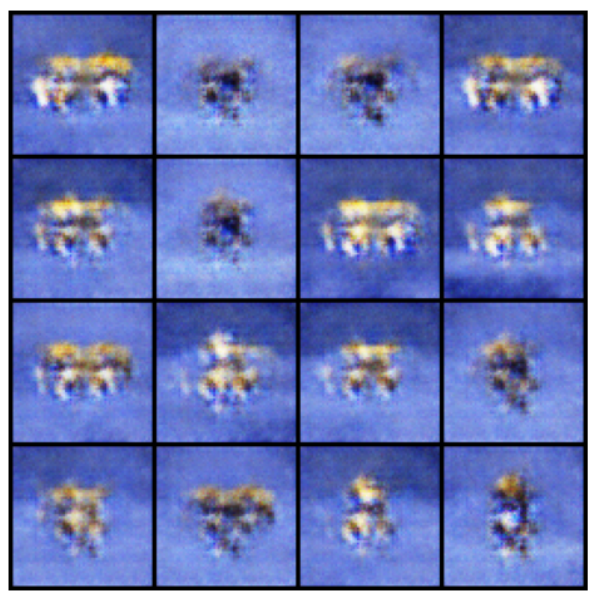

Fig. 2: Best synthetic samples produced by the trained DCGAN. The model captures overall body shape and silhouette, but generated images are blurry and lack fine detail, reflecting the difficulty of learning a complex distribution from only 82 training images.

Although visually recognisable animals were produced, appending synthetic images to the real training set did not consistently improve classifier accuracy. GANs generally require thousands of samples to learn high-fidelity distributions;

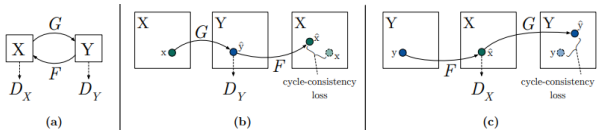

Fig. 3: CycleGAN model architecture.

with 82 images, mode collapse and blurry outputs degraded the effective data quality.

VI. CycleGAN for Unpaired Style Transfer

A. Motivation and Formulation

CycleGAN [4] performs unpaired image-to-image translation between two domains X (cows) and Y (horses) without requiring pixel-aligned pairs. Two generators and two discriminators are trained jointly:

$$\begin{aligned} G &: X \to Y \qquad \text{(cow} \to \text{horse)} \\ F &: Y \to X \qquad \text{(horse} \to \text{cow)} \\ D_Y &: \text{real vs. fake horses} \\ D_X &: \text{real vs. fake cows} \end{aligned}$$

B. Loss Components

a) Adversarial loss: (for G; analogous for F):

$$\mathcal{L}_{\text{GAN}}(G, D_Y, X, Y) = \mathbb{E}[\log D_Y(y)] + \mathbb{E}[\log(1 - D_Y(G(x)))] \tag{7}$$

b) Cycle-consistency loss: — ensures round-trip fidelity to prevent the mappings from collapsing:

$$\mathcal{L}_{\text{cyc}}(G, F) = \mathbb{E}[\|F(G(x)) - x\|_1] + \mathbb{E}[\|G(F(y)) - y\|_1] \tag{8}$$

c) Identity loss: — preserves colour and texture when the input already belongs to the target domain:

$$\mathcal{L}_{\text{id}}(G, F) = \mathbb{E}[\|G(y) - y\|_1] + \mathbb{E}[\|F(x) - x\|_1] \tag{9}$$

d) Full objective::

$$\mathcal{L} = \mathcal{L}_{\text{GAN}}(G, D_Y) + \mathcal{L}_{\text{GAN}}(F, D_X) + \lambda_{\text{cyc}}\, \mathcal{L}_{\text{cyc}} + \lambda_{\text{id}}\, \mathcal{L}_{\text{id}} \tag{10}$$

with $\lambda_{\text{cyc}} = 10$ and $\lambda_{\text{id}} = 5$ following the original paper.

C. Training Behaviour and Results

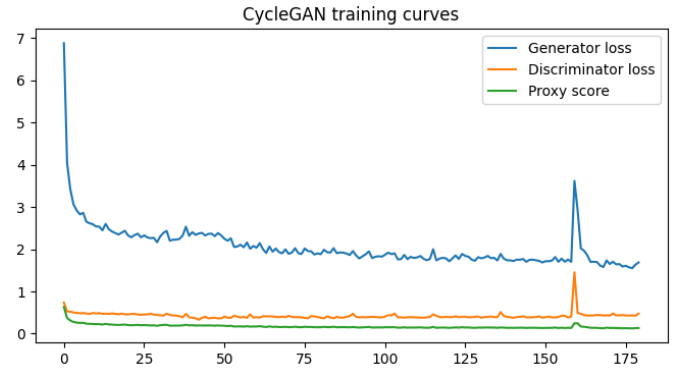

Fig. 4: CycleGAN training loss curves. Both generator and discriminator losses decrease smoothly, indicating a more stable adversarial balance than DCGAN — aided by cycle-consistency and identity constraints.

Fig. 5: CycleGAN translation results: Cow → Horse (top row) and Horse → Cow (bottom row). The model successfully transfers coat texture and colour while preserving body pose, scene layout, and background content. Minor texture artefacts remain near the animal boundaries.

CycleGAN converged significantly more stably than DCGAN. Because the generator receives a full structural image rather than random noise, it only needs to learn *appearance transfer*, a substantially easier task than unconditional generation — particularly under data scarcity.

D. CycleGAN Images as a Classifier Training Source

As a final experiment, the 80 translated images produced by the trained CycleGAN generators (40 Cow→Horse and 40

Horse→Cow) were appended to the 82 real training images, forming a combined set of 162 samples. A fresh ResNet18 was then trained on this combined dataset using the same two-stage fine-tuning protocol described in Section IV.

Despite reaching a training accuracy of 99.4% in Stage B, the model achieved only **68.29%** test accuracy — a drop of more than 23 percentage points compared to the real-data-only baseline (91.46%). Per-class metrics are shown in Table II.

TABLE II: Per-class metrics for the CycleGAN-augmented classifier.

Class	Precision	Recall	F1
Cow	0.64	0.85	0.73
Horse	0.78	0.51	0.62
Macro avg.	**0.71**	**0.68**	**0.67**

The large gap between training and test accuracy indicates severe overfitting induced by the synthetic images. Three factors explain this outcome:

1) **Domain shift.** CycleGAN-translated images were generated at 128×128 resolution and up-sampled to 224×224 for ResNet18, introducing interpolation artefacts absent from the real images.
2) **Texture bias.** The translated images share texture characteristics inherited from the source domain (e.g., a cow-shaped silhouette with horse colouring), which introduces ambiguous or misleading features that confuse the classifier.
3) **Distribution mismatch.** With only 82 real images, the CycleGAN generators themselves are imperfectly trained; the resulting translated images do not faithfully represent the target class distribution, effectively adding label noise.

This result is consistent with the general finding that GAN-based augmentation requires high-fidelity generation quality — a condition that is difficult to meet with fewer than 100 training images per class.

VII. Comparison: DCGAN vs. CycleGAN

TABLE III: Architectural and behavioural comparison of the two GAN models.

Property	DCGAN	CycleGAN
Input	Noise $z \sim p_z$	Real image x
Output	Synthesised image	Domain-translated image
Generators	1	2 (G and F)
Discriminators	1	2 (D_X and D_Y)
Auxiliary losses	None	Cycle + Identity
Small-data suitability	Low	Moderate–High
Training stability	Unstable	Stable
Main use case	Unconditional gen.	Style / domain transfer

VIII. Experimental Results

A. Classification Accuracy

TABLE IV: Test accuracy across all experiments.

Experiment	Test Accuracy
Baseline CNN (from scratch)	55–70%
Regularised CNN (dropout + L2)	≈73%
DCGAN augmentation (best run)	75.61%
Transfer learning (final)	**91.46%**
CycleGAN augmentation + Transfer learning	68.29%

B. Per-Class Metrics — Final Classifier (Real Images Only)

TABLE V: Precision, recall, and F1-score for the best transfer learning model (real images only).

Class	Precision	Recall	F1-score
Cow	0.85	1.00	0.92
Horse	1.00	0.83	0.91
Macro avg.	**0.93**	**0.92**	**0.92**

The classifier shows high precision for horses (no false positives) and perfect recall for cows (no missed detections), resulting in balanced F1-scores above 0.91 for both classes.

IX. Discussion

Several key findings emerge from our experiments:

1) **Transfer learning is the dominant strategy.** Pretrained ImageNet features contain rich, generalisable representations that dramatically reduce the data requirements for downstream tasks. The two-stage fine-tuning protocol was critical: Stage A prevented head initialisation from corrupting backbone features.
2) **DCGAN is limited by dataset size.** Adversarial training from noise requires thousands of images to learn a stable generator distribution. With 82 images per class, generated samples were blurry and introduced noise rather than useful diversity.
3) **CycleGAN benefits from structural priors.** By conditioning on existing images rather than random noise, CycleGAN circumvents the hardest aspects of unconditional generation, making it far better suited to small-dataset regimes for visual style transfer.
4) **CycleGAN augmentation hurt classifier accuracy.** Appending translated images reduced test accuracy from 91.46% to 68.29%. The translated images introduced domain-shift artefacts, texture ambiguity, and label noise that outweighed any benefit from increased dataset size. This is consistent with the broader finding that GAN-based augmentation requires near-photorealistic generation quality to benefit downstream classifiers.
5) **Data quality outweighs synthetic quantity.** Appending low-quality GAN samples — whether from DCGAN or CycleGAN — hurt classifier performance; using only real

images with geometric augmentation was consistently superior.

6) **Generative and discriminative goals are complementary but separate.** Transfer learning handles classification; GAN models handle style transfer and domain visualisation. The two objectives should not be conflated: strong visual translation does not imply useful augmentation for classification.

X. Conclusion and Future Work

This paper presented a systematic study of binary animal image classification under severe data scarcity. Starting from a baseline CNN achieving 55–70%, we progressively improved to **91.46%** through transfer learning with ImageNet-pretrained features. DCGAN demonstrated the challenges of generative modelling on small datasets, while CycleGAN proved highly effective for unpaired style transfer, successfully converting cow images to horse images and vice versa while preserving spatial structure. A final experiment using CycleGAN-translated images as a data-augmentation source confirmed that visual translation quality alone does not guarantee classifier improvement: test accuracy fell to 68.29%, underscoring that synthetic diversity must be accompanied by high fidelity to benefit discriminative models.

Future directions include:

- Collecting more annotated data or leveraging semi-supervised learning.
- Replacing DCGAN with WGAN-GP [1] to stabilise training.
- Exploring diffusion models as an alternative generative backbone, which may produce higher-fidelity images at small dataset sizes.
- Applying quality filtering to synthetic images before augmentation, discarding samples with high artefact scores.
- Applying k-fold cross-validation for more robust performance estimates.

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